



Piscine Pro AI / Machine Learning

Simple Linear Regression

Summary: In this Module, you will learn about Simple Linear Regression.

Version: 1.1

Contents

I	Introduction	2
II	General Instructions	3
III	Exercise 00	4
IV	Exercise 01	5
V	Exercise 02	6
VI	Submission and peer-evaluation	7

Chapter I

Introduction

Welcome !



If you haven't already done so, read `en.toolkit.pdf`.

What this Module will show you:

In this module, you will learn about the practical application of simple linear regression through hands-on experience with essential libraries and tools. We will explore the utilization of programming libraries like pandas for efficient data manipulation, matplotlib for visualizing results, and potentially specialized tools like scikit-learn to streamline the implementation of regression models. This practical exposure will equip you with the skills needed to effectively apply simple linear regression to real-world datasets, bridging the gap between theoretical understanding and the technical proficiency required for successful data analysis.

Good luck to all.

Chapter II

General Instructions

Unless explicitly stated otherwise, the following rules apply every day of this Piscine Pro.

- Assignments must be completed in the specified order. Subsequent assignments will only be assessed if all previous ones have been correctly completed.
- Your assignments will be reviewed by your peers.
- You must not include any files in your submission other than those explicitly requested by the assignments.
- If you have a question, ask your left neighbor first. If that doesn't help, try your right neighbor.
- Any technical information you may need can be found on the Internet.

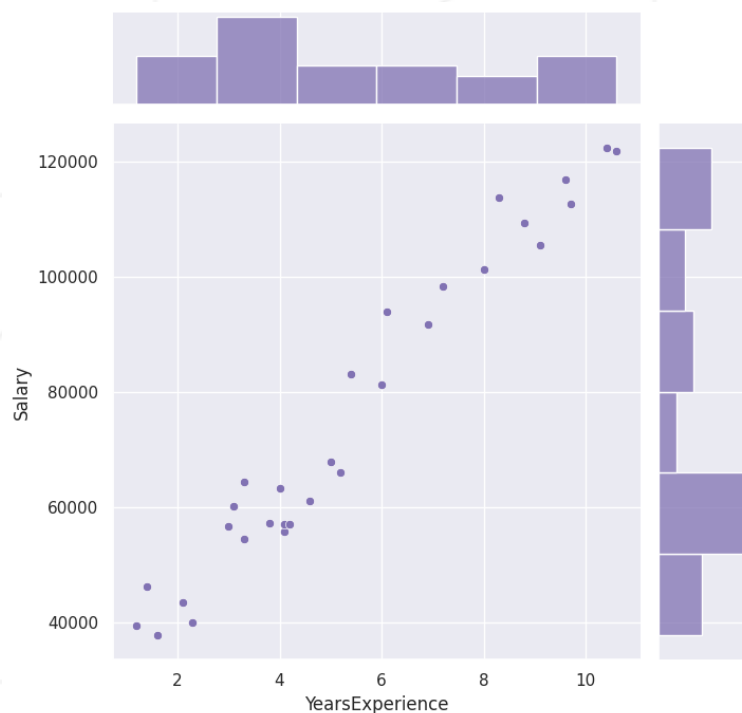
Chapter III

Exercise 00

	Exercise 00
Displaying your data	
Turn-in directory: <i>ex00/</i>	
Files to turn in: Beginner00.ipynb	
Allowed functions: pandas, seaborn	

For this first exercise, you'll need to load the data into colab and display it successfully in a graph with the distribution, you'll need to use the pandas or seaborn library.

You should have something like this:



Chapter IV

Exercise 01

	Exercise 01
Train and Display	
Turn-in directory: <i>ex01/</i>	
Files to turn in: Beginner00.ipynb	
Allowed functions: pandas, seaborn, sklearn	

For this second exercise we're going to get down to business: you'll have to train your first model.

You should then display your linear regression bar on your first graph.

You should have something like this:



Chapter V

Exercise 02

	Exercise 02
Predict	
Turn-in directory: <i>ex02/</i>	
Files to turn in: Beginner00.ipynb	
Allowed functions: sklearn	

Great, now you're going to provide a salary based on the number of years of experience.

You should find a result close to this one with 10 years of experience:

```
Predicted salary for 10 years of experience : 119347.82718107398
```

Chapter VI

Submission and peer-evaluation

- Create a `professional_training_beginner` folder at the root of your home, and move around in it.
- Create a new `module00` folder and navigate to it.